Power Supply

12 V DC Source with 3.3 V Logic Regulation Stage

Confirmed: 12 V Battery (BAT1) | Assumed: AMS1117-3.3 LDO Regulator

Identification note: *The schematic (ROOT.DSN) confirms a multi-cell* ***BATTERY*** *component labelled* ***BAT1*** *supplying* ***12 V****, which powers the motor/MOSFET H-Bridge stage directly. The schematic also contains a net labelled* ***3V3****, indicating a regulated 3.3 V logic rail for the STM32F103C6 microcontroller; however, no specific voltage-regulator IC part number appears in the captured schematic data (it is likely provided by an onboard regulator on the microcontroller board/module rather than being drawn as a discrete part). The* ***AMS1117-3.3*** *LDO regulator is documented below as the closest commonly-used equivalent for this function, and is clearly marked as an assumed component.*

1. Component Name

Element	Part / Value	Status
Primary DC Source	12 V Battery Pack (BAT1)	Confirmed in schematic
Logic Regulator	AMS1117-3.3 (3.3 V, 1 A LDO)	Assumed closest common equivalent

2. Introduction

The power supply section provides the electrical energy required by every other stage of the circuit. In this project it consists of two levels: a primary 12 V DC source (a multi-cell battery) that directly powers the H-Bridge/motor stage, and a secondary regulated 3.3 V rail that supplies clean, stable logic-level power to the STM32F103C6 microcontroller and any associated digital circuitry.

3. Manufacturer

Element	Manufacturer	Official Datasheet
12 V Battery (BAT1)	Generic multi-cell battery pack (manufacturer not specified in schematic)	N/A — generic power source symbol
AMS1117-3.3 Regulator	Advanced Monolithic Systems (also second-sourced by many manufacturers)	AMS1117 1A Low Dropout Voltage Regulator Datasheet

4. Functional Description

Working Principle

The 12 V battery pack provides raw DC power directly to the H-Bridge, whose MOSFETs switch this voltage across the DC motor. Because the STM32 microcontroller's logic operates at 3.3 V, a linear low-dropout (LDO) regulator such as the AMS1117-3.3 is used to step this down (from the battery voltage or from an intermediate 5 V rail) to a clean, stable 3.3 V supply. An LDO regulator maintains its output voltage constant by continuously adjusting the conduction of an internal pass transistor to compensate for changes in input voltage or load current, dissipating the voltage difference as heat.

Purpose / Why it is used in this H-Bridge DC Motor Driver with PWM project

This project requires two different voltage domains: a higher-voltage, higher-current supply for the motor and switching MOSFETs, and a clean, low-noise low-voltage supply for the microcontroller that generates the PWM control signals. Separating these supplies (while sharing a common ground) protects the sensitive digital logic from switching noise and voltage transients generated by the motor and MOSFET switching.

5. Key Features

Feature	12 V Battery Source	AMS1117-3.3 Regulator
Output Voltage	12 V DC (nominal)	3.3 V DC (fixed)
Output Current	Application dependent (multi-cell pack)	Up to 1 A
Dropout Voltage	N/A	1.3 V (typical, max at 800 mA-1A)
Package Type	N/A (battery pack)	SOT-223 / TO-252 / SO-8
Operating Temperature	Typical commercial range	0 degC to 125 degC (junction, typical)
Protection Features	N/A (external fusing recommended)	Thermal shutdown, short-circuit current limiting
Efficiency	N/A (direct source)	Linear regulator; efficiency approx. Vout/Vin (moderate)

6. Pin Configuration

Battery Pack (BAT1)

Terminal	Name	Function
1	+ (Positive)	12 V positive supply output
2	- (Negative/GND)	Common ground / return path

AMS1117-3.3 (SOT-223, assumed component)

Pin Number	Pin Name	Function
1	GND/ADJ	Ground reference (fixed-voltage version)
2	VOUT	Regulated 3.3 V output
3	VIN	Unregulated input voltage
4 (tab)	VOUT (tab)	Thermal tab, internally connected to VOUT

7. Electrical Specifications

Parameter	Symbol	Value	Conditions
Battery Output Voltage	V	12 V DC	Nominal
Regulator Output Voltage	VOUT	3.3 V (fixed)	AMS1117-3.3
Regulator Max. Output Current	IOUT	1 A	-
Regulator Dropout Voltage	VDROP	1.3 V (max, typ.)	At 800 mA-1A load
Regulator Max. Input Voltage	VIN(max)	15 V	-
Line Regulation	-	0.2% (max)	-
Load Regulation	-	0.4% (max)	-

8. Typical Applications

- Powering microcontroller development boards and embedded systems
- Battery-powered portable electronics
- Motor driver and H-Bridge power stages
- Sensor and analog front-end power rails
- IoT and single-board computer power regulation
- USB-powered device 5V-to-3.3V conversion
- General-purpose prototyping and evaluation boards
- Powering logic-level ICs from a higher unregulated voltage source

9. Advantages

- Simple, low-component-count regulated supply design
- Low dropout voltage allows efficient regulation from moderately higher input voltages
- Built-in thermal shutdown and current-limiting protect against overload and overheating
- Battery source provides portable, isolated power independent of mains supply
- Low cost and wide second-source availability for the regulator

10. Limitations

- Linear regulator dissipates excess voltage as heat, reducing efficiency at higher input-output differentials
- Not suitable for stepping down large voltage differences at high current (e.g., 12 V to 3.3 V at high load) without excessive heat generation
- Battery capacity is finite and requires periodic recharging or replacement
- No inherent voltage step-up capability (only step-down/regulation)
- Requires adequate output capacitance for stability, as specified in the regulator datasheet

11. Role in This Project

The 12 V battery is the primary energy source for the entire circuit, directly powering the H-Bridge MOSFETs and, through them, the DC motor. A separate regulated 3.3 V rail (assumed to be derived via an AMS1117-3.3 or an equivalent onboard LDO regulator on the microcontroller module) powers the STM32F103C6 microcontroller, ensuring it receives a clean, stable logic supply isolated from the electrical noise generated by motor switching, so that PWM generation and control logic remain reliable.

12. Precautions

- Fuse the battery supply line to protect against short-circuit or overcurrent faults
- Observe correct battery polarity when connecting to the circuit
- Add adequate input and output decoupling/bulk capacitors around the voltage regulator, as recommended in its datasheet
- Ensure the regulator has sufficient heat-sinking/copper area if operating near its maximum current or voltage differential
- Keep high-current motor return paths separate from sensitive logic ground where practical to reduce noise coupling
- Do not exceed the regulator's maximum input voltage rating

13. Official Reference

Official Manufacturer	Advanced Monolithic Systems (AMS1117 family; battery pack manufacturer not specified in schematic)
Official Datasheet Title	AMS1117 1A Low Dropout Voltage Regulator Datasheet
Datasheet Version	Manufacturer standard datasheet (AMS1117-3.3 fixed-output variant)
Component Status	Battery (BAT1): confirmed in schematic. Regulator: assumed closest common equivalent